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Shoot(f) for the Stars: Calculating the ZAMS Interior Structure of a $2M_\odot$, Solar Composition Star

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1. THE PHYSICAL PROBLEM

The interior structure of stars can be solved numerically with a set of four coupled differential equations in luminosity, temperature, pressure, radius, and mass, with either mass or radius acting as the independent variable. Such calculations are evaluated over a range of stellar masses and compositions and may be used to explain observable properties of stars. For example, these calculations can determine core temperatures and pressures for stars of different masses, which can reveal the source of fusion (primarily CNO vs PP chain) within the star. Such calculations can determine which regions in a star are radiative or convective, and the switching of convective to radiative (in stars more massive than 1.5 $M_\odot$) defines the core/outer layer boundary. The core/outer layer boundary is the site of magnetic field generation due to shear between the two layers, where mixing between the two layers occurs (resulting in lithium depletion in the photosphere, for example). In astroseismology, p waves (in the radiative outer layers) begin to dominate over g waves (in the convective core), so being able to differentiate the regions with detailed modeling is important to interpreting astroseismic signals. Therefore, accurate models of the inner structure of stars are essential to our understanding of the internal and observable surface processes in stars.

In this work, I calculate the interior structure of a 2 $M_\odot$ star assuming Lagrangian coordinates (mass as independent variable) to determine how each of the four remaining parameters vary throughout the star using equations from C. J. Hansen et al. (2004) and energy generation rate approximations from R. Kippenhahn et al. (2012). In section 2, I describe the physical setup of the star, including section 2.1, which outlines how I choose inner (core) and outer (surface) boundary conditions for the integration, and section 2.2, where I describe how each parameter varies over changing mass. In section 3, I describe the procedure I implement to numerically solve the equations of stellar structure. In section 3.1, I outline how I modeled a 2 $M_\odot$ star using the aforementioned approach, and comment on the efficiency of my model. Finally, I compare my calculated stellar structure with that of an identical star modeled with `MESA` (B. Paxton et al. 2011) in section 3.2

2. PARAMETERS USED

The four coupled differential equations of stellar structure depend on mass (m, in g), radius (R, in cm), luminosity (L, in ergs), temperature (T, in K), and pressure (P, in dynes/cm2), where each variable has additional dependencies, such as on density (ρ, in g/cm3), energy generation rate (ϵ, in ergs/g), and opacity (κ, in cm2/g). Mass and composition are independent variables which the luminosity, temperature, radius, and pressure vary over.

2.1. *Initial boundary conditions*

At the surface of the star, the initial guesses are the luminosity of the star and the radius at the surface, which can be derived from stellar homology relations for a certain mass star (equations 1 and 2). For massive stars ($> 1\ M_\odot$),

$$\frac{R_*}{R_\odot} \propto (\frac{M_*}{M_\odot})^{0.75} \tag{1}$$

$$\frac{L_*}{L_\odot} \propto (\frac{M_*}{M_\odot})^{3.5} \tag{2}$$

From an initial total luminosity L_* and stellar radius R_*, the surface temperature $T_{\mathrm{eff},*}$ can be calculated using the Stefan-Boltzmann law. The outer pressure condition is given by two coupled equations; one pertaining to the pressure from gravity (eq 3), the other pertaining to the ideal gas and radiation pressure equations of state (eq 4).

$$P_{\mathrm{grav}} = \frac{2}{3\kappa(\rho, T)} \frac{GM_*}{R_*^2} \tag{3}$$

and

$$P_{\mathrm{gas+rad}} = \frac{\rho N_A k T}{\mu(X, Y, Z)} + \frac{1}{3} a T_{\mathrm{eff},*}^4 \tag{4}$$

To find the pressure at the surface of the star, the two equations are simultaneously solved as a function of density ρ_*. Once the density at the surface of the star

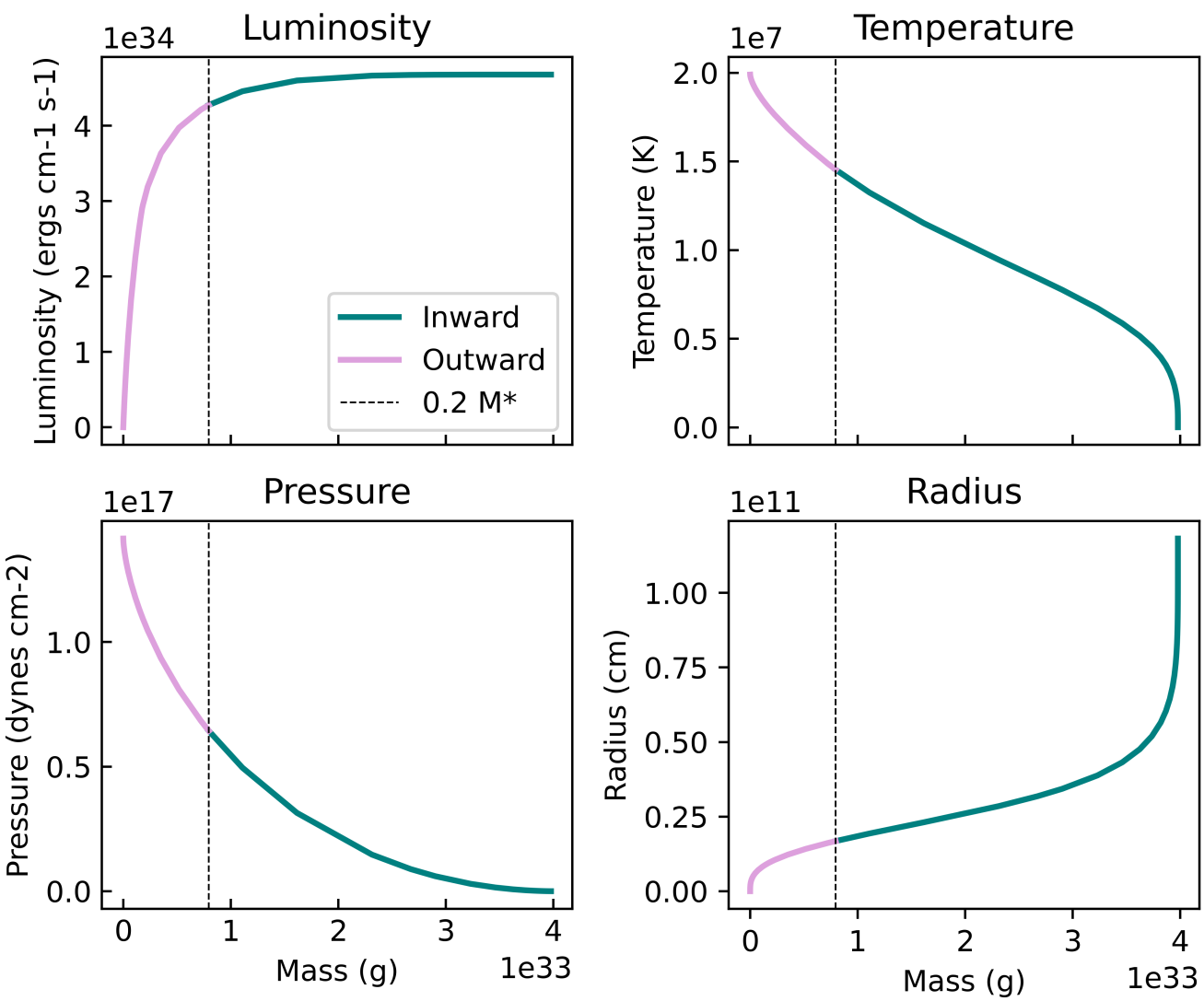

Figure 1. Converged stellar structure calculation for a 2 $M_\odot$ star. Integration from the surface inwards to 0.2 M_* is represented in teal, while the integration from the core outwards to 0.2 M_* is plotted in pink. The 0.2 M_* boundary is shown as a dashed line.

ρ_* is found, the surface pressure P_* is calculated given that density and either equation for pressure.

The core of the star is defined just exterior to the $M_{\text{enclosed}} = 0$ point, as defining the initial inner conditions at $M_{\text{enclosed}} = 0$ would lead to discontinuities in the calculations. I select my core mass to start at 10^{10} g, which is very small relative to the total mass of the star (order $> 10^{33}$g)

At the core of the star, I provide initial guesses for the core temperature and core pressure. From these, the radius of the core is calculated given the density of the core ρ_c, which is found given the initial temperature and pressure guess (eq 5). The luminosity at the core is a function of the energy generation rate given the core temperature and density (eq 6; described further in section 2.2).

$$R_c = (\frac{3}{4\pi\rho_c})^{1/3} m_{\text{enclosed}}^{1/3} \tag{5}$$

$$L_c = \epsilon m_{\text{enclosed}} \tag{6}$$

The pressure (P_r) a small mass step out from the core (m_{enclosed}) decreases from the core pressure P_c guess by a small fraction depending on the density, given by eq. 7:

$$P_r = P_c - \frac{3G}{8\pi}(\frac{4\pi}{3\rho_c})^{4/3} m_{\text{enclosed}}^{2/3} \tag{7}$$

The temperature in the core depends on if the core is radiative or adiabatic. Since I am assuming a massive star, the core is expected to be convective. To further check this, the radiative and convective temperatures (eqs. 9, 11) and their associated gradients $\nabla_{\rm rad}$ and $\nabla_{\rm ad}$ are calculated (eqs.8 and 10). If the radiative temperature gradient is greater than the adiabatic temperature gradient, the region is convective, and if the adiabatic gradient is larger, the region is radiative.

For radiative,

$$\nabla_{\rm rad} = \frac{3}{16\pi ac}\frac{P\kappa}{T_{\rm rad}^4}\frac{L}{Gm_{\rm enclosed}} \tag{8}$$

$$T_{\rm rad} = (T^4 - (\frac{1}{2ac})(\frac{3}{4\pi})^{2/3}\kappa\epsilon\rho^{4/3}m_{\rm enclosed}^{2/3})^{1/4} \tag{9}$$

For convective regions,

$$\nabla_{\rm ad} = 1 - \frac{1}{\gamma}; \gamma = 5/3 \tag{10}$$

$$T_{\rm conv} = \log(T) - (\frac{\pi}{6})^{1/3}\frac{G\nabla_{\rm ad}\rho^{4/3}}{P}m_{\rm enclosed}^{2/3} \tag{11}$$

2.2. *Derivatives*

Now that the initial boundary conditions have been set, I discuss how each of the parameters evolves over changing mass.

The luminosity of the star at a certain mass coordinate varies depending on the energy generation rate of the star in that mass shell, which is a sensitive function of temperature and density (eq. 12). The energy generation rate is a combination of the energy generated from the proton-proton (PP) chain and from the carbon-nitrogen-oxygen (CNO) cycle. Both of these cycles have a strong temperature dependence, with the generation rate of the CNO cycle increasing more strongly in temperature ($\epsilon \propto T^{15}$) than the PP chain ($\epsilon \propto T^4$). Therefore, the CNO cycle dominates in the hottest parts of the core, whereas the PP chain dominates in the cooler outer portions of the core. Once the temperature drops below $\sim 10^7$, the energy generation rate slows significantly, indicating the edge of the core as well as the transition of the star from the convective core to the radiative outer layers. In this work, I adopt energy generation rate approximations from (R. Kippenhahn et al. 2012).

$$\frac{dL}{dm} = \epsilon \tag{12}$$

The temperature of the star varies as a function of the mass of the layer, current temperature, pressure, and the radius corresponding to an enclosed mass (eq. 13). The mechanism of heat transport (radiative or convective) additionally governs the rate at which temperature changes. For convective regions, the temperature varies based on the adiabatic gradient $\nabla_{\rm ad}$, which is only a function of the adiabatic exponent γ=5/3. In radiative regions, the radiative gradient $\nabla_{\rm rad}$ is instead applied, which depends on the pressure, luminosity, radius, and opacity, as described in section 2.1.

$$\frac{dT}{dm} = \frac{GmT}{4\pi R^4 P}\nabla \tag{13}$$

I calculate opacity of the star, κ by interpolating a table of precomputed opacities assuming fully ionized material of a given composition X (hydrogen fraction), Y (helium fraction), Z (metal fraction). The opacity at a mass coordinate in the star depends on the temperature and density of the region, where the density depends additionally on the composition, temperature, and pressure. Since the opacity table is calculated assuming complete ionization, this sets a lower limit on the mass of star that can be calculated. For lower mass stars, outer regions of the star might not be hot enough to be fully ionized, which means that the opacities interpolated from the table do not apply. The opacity table sets a lower limit on temperature at log(T)=3.75, corresponding to a temperature of 5623 K. To avoid running into interpolation issues, I limit my calculation to stars more massive than the Sun, which has a $T_{\rm eff}$ = 5778 K (at $\tau \sim 2/3$).

The change in pressure as a function of mass is just a function of the radius of a layer (eq. 14). The change in radius depends on the density and current radius (eq. 15).

For pressure,

$$\frac{dP}{dm} = \frac{-Gm}{4\pi R^4} \tag{14}$$

and for radius,

$$\frac{dR}{dm} = \frac{1}{4\pi\rho R^2} \tag{15}$$

3. NUMERICAL METHODS

When allowing mass as the independent variable and selecting a composition in the form of hydrogen fraction, helium fraction, and metal fraction (X, Y, Z), a star can be constructed by marching the mass coordinate both inwards and outwards and determining how each of the four parameters vary. In this work, I apply the "shooting for a fixed point" method, which involves initializing inner and outer boundary conditions at reasonable guesses for each of the four parameters

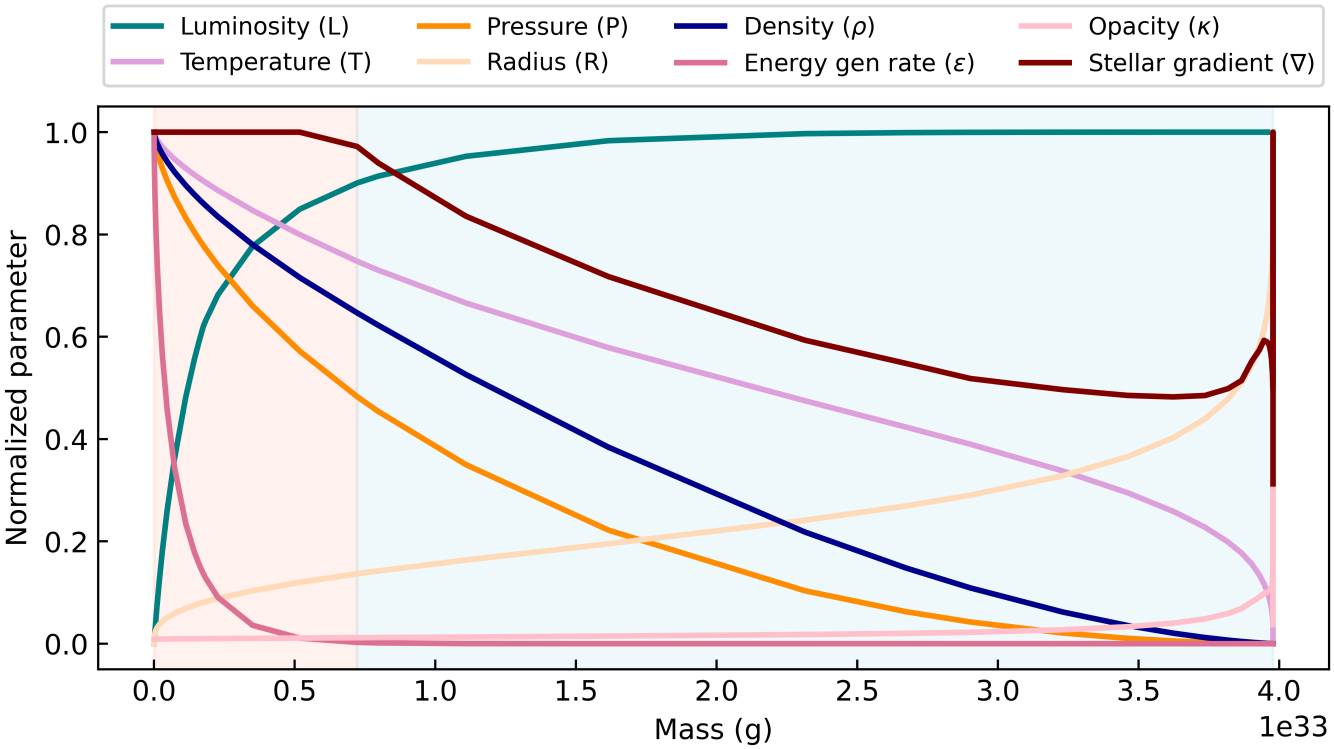

Figure 2. Normalized parameters corresponding to the columns in Table 3.2. Luminosity, pressure, temperature, and radius are output from the stellar structure integration, whereas the density, energy generation rate, opacity, and the stellar gradient are calculated from the converged luminosity, pressure, temperature, and radius. At the surface of the star (Mass = 4×10^{33} g, the pressure and density get very small, which causes the opacity and stellar gradient calculations to experience numerical errors.

and marching the inner guesses outwards and the outer guesses inwards until they meet in the middle. The "marching" is done by increasing the Lagrangian mass coordinate simultaneously outwards from the center and inwards from the surface and varying each of the dependent variables (luminosity, temperature, radius, pressure) by its associated mass derivative. To do so, I use the `scipy.integrate.solve_ivp` function, which takes as inputs initial conditions, a function that determines how these conditions vary, and the bounds of the input variable domain over which to vary the initial conditions using the input function. Here, I use `scipy.integrate.solve_ivp` to integrate the initial core guess outwards using my `derivs` function, which returns the derivatives of luminosity, radius, pressure, and temperature as a function of mass. The mass range input goes from a small core mass (representing somewhere very close to, but not at, the center of the star to avoid singularities) to 0.2 times the mass of the star. My initial guesses at the core are calculated as described above using my `load1` function. I run the same calculation from the outer surface in to the 0.2 M_* point using the same `derivs` function and initial conditions set by the `load2` function.

Once the inwards and outwards integrations meet in the middle at a mass point selected to be 0.2 times the total mass of the star M_*, the convergence is checked by determining how well the inwards-out and outwards-in predictions for each of the four dependent variables compare. To ensure that the two solutions converge to the same values in luminosity, temperature, radius, and pressure at 0.2 M_*, the integrations are included within a function that minimizes over the initial guesses; `scipy.minimize.fsolve`. This function takes as input a function called `difference`, which returns the difference between the current guess at the 0.2 M_* boundary scaled by each parameter's initial guess, as well as a set of initial guesses for the boundary conditions P_c, T_c, R_*, and L_*.

The final solution to the stellar structure calculation returns initial guesses for the inner and outer boundary conditions that result in converged luminosity, temperature, radius, and pressure at a mass of 0.2 M_*. Then, using the preferred inner and outer boundary conditions, the complete structure of the star in luminosity, temperature, radius, and pressure as a function of mass is evaluated by running the integration inwards and outwards once more.

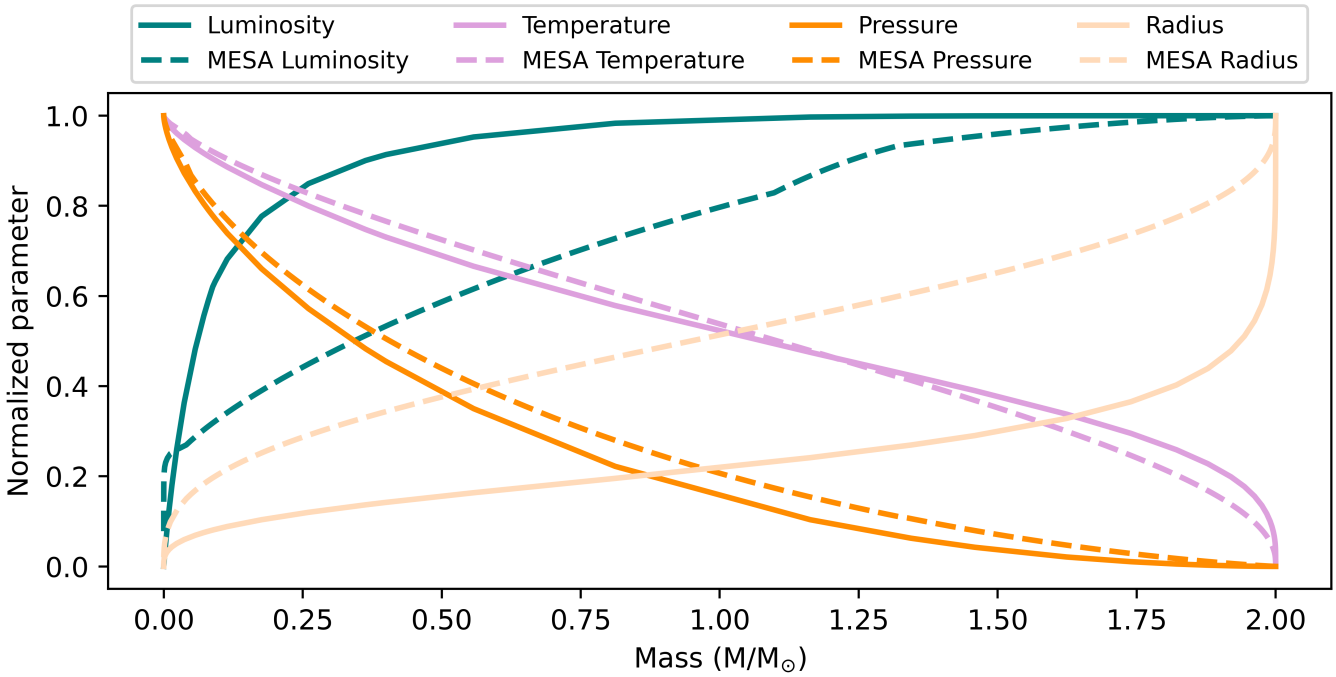

Figure 3. Comparison between the stellar structure model calculated as described in this work (solid lines) and the Modules for Experiments in Stellar Astrophysics (`MESA`) code (dashed lines). Both models have each parameter normalized by the maximum value for comparison. I note that in reality, the `MESA` model and my model do not align in their inner and outer boundary conditions, with the boundary condition values displayed in Table 3.1.

	`MESA`	This work	Difference
Core			
M_c (g)	3.10E+26	3.10E+26	0.00
L_c (ergs)	7.30E+27	1.20E+29	-15.44
T_c (K)	3.30E+06	2.00E+07	-5.06
P_c (dynes/cm2)	7.40E+13	1.40E+17	-1,890.89
R_c (cm)	7.60E+08	1.10E+08	0.86
Surface			
M_s (g)	4.00E+33	3.98E+33	0.01
L_s (ergs)	4.60E+34	4.70E+34	-0.02
T_s (K)	5.00E+03	8.30E+03	-0.66
P_s (dynes/cm2)	1.80E+04	3.90E+03	0.78
R_s (cm)	3.30E+11	1.20E+11	0.64

Table 1. Core and surface values of my model compared to the `MESA` model, where the difference column corresponds to the the difference between the `MESA` model and my model scaled by the `MESA` model value. For direct comparison of the core values, I evaluated my model at a core mass matching the `MESA` core mass (3.1×10^{26} g) rather than the core mass I assumed throughout (10^{10} g)

3.1. *Modeling stars of different masses*

Since I am limited to modeling massive stars by the opacity table requiring complete ionization, I first select a 2 $M_\odot$ star to model. I assume this star has a solar composition, with X = 0.70, Y = 0.28, and Z = 0.02. Given the homology relations for massive stars, I set my stellar radius guess at 1.17×10^{11} cm and stellar luminosity to 4.33×10^{34} ergs. I allow the initial core pressure and temperature to be similar to those for the Sun, at $P_c = 1.5 \times 10^{17}$ dynes/cm^2 and $T_c = 2 \times 10^7$ K, as I find my convergence is not as dependent on the inner boundary guesses as it is on the outer boundary guesses.

The quality of the model is assessed by the number of iterations required to reach a converged solution. My model of a 2 $M_\odot$ star converges after 52 iterations, indicating that some improvements might be made to ensure the model converges with less iterations. Possible improvements include using better initial guesses. It is also possible that I have made a small calculation error somewhere, which might throw off the integrator `scipy.integrate.fsolve` when searching for a converged solution

I also tested modeling stars of different masses, assuming the same initial P_c and T_c guesses, but varying the L_* and R_* initial guesses by the massive star homology relations. I find I can reliably model stars with masses between 1.5 and 2 $M_\odot$, assuming a solar composition. The 1.5 solar mass star converged after 37 iterations, which is the best convergence I am able to achieve.

3.2. *Modeling a 2 $M_\odot$ star with* `MESA`

To compare how well my models of a star's structure compare to the widely used stellar modeling software `MESA` (B. Paxton et al. 2011), I set up and model a 2 $M_\odot$, solar composition star using `MESA`. In short, `MESA` initializes and evolves a star of a given mass and composition from the pre-main sequence to a given age, and outputs both the evolutionary history of the star and its internal structure. I can then compare the inner structure of a ZAMs star computed with MESA to the structure I numerically solved for in this work to determine how accurate my approximations of stellar structure are. To simulate a star with `MESA`, I edit the `inlist_project` file to correspond to a 2 $M_\odot$ star with a metal fraction of 0.02 and a helium fraction of 0.28, as well as edit the profile configuration file to include luminosity as a model output. I then run the star calculation, and plot the outputs alongside my outputs in figure 3.

I find that on an order of magnitude basis, my calculations tend to agree with the calculations from `MESA`, shown in table 3.1. The parameters match much closer at the surface of the star than at the core. The radius and luminosity calculated by `MESA` increase more gradually across increasing mass than my calculations, which implies I might be incorrectly calculating these parameters within my model, or that the more complex `MESA` star model is accounting for more physics, ex. has more opacities included or different approximations for energy generation rates than my model. My core pressure agrees the least with the `MESA` model, with my model being larger by a factor of nearly 2000. However, the core pressure of a star more massive than the Sun should be greater than the core pressure of the Sun (order 10^{17} dynes/cm2), which both my model and the `MESA` model do not well predict. My model is closer in core pressure to the Sun than the `MESA` model, which makes me doubt the accuracy of the rest of the `MESA` model. When just comparing the normalized parameters, as in Figure 3, I find that the pressure and temperature change at a similar rate from their core to surface values compared to my models, whereas the radius and luminosity vary differently.

M_{enc}	L	T	P	R	ρ	ϵ	Transport	κ	∇_{ad}	∇
g	ergs	K	dynes/cm^2	cm	g/cm^3	ergs/g	-	cm^2/g	-	-
3.98e+33	4.68e+34	8.28e+03	3.87e+03	1.18e+11	3.47e-09	2.84e-74	radiative	3.27	0.4	0.125
3.98e+33	4.68e+34	8.74e+03	5.16e+03	1.18e+11	4.37e-09	2.84e-74	radiative	6.95	0.4	0.285
3.98e+33	4.68e+34	1.64e+04	2.54e+04	1.18e+11	1.14e-08	2.84e-74	convective	59.1	0.4	0.4
3.98e+33	4.68e+34	3.05e+04	1.52e+05	1.18e+11	3.63e-08	2.84e-74	radiative	35.5	0.4	0.287
3.98e+33	4.68e+34	5.82e+04	1.38e+06	1.17e+11	1.72e-07	2.84e-74	radiative	42.9	0.4	0.239
3.98e+33	4.68e+34	9.36e+04	1.17e+07	1.16e+11	9.16e-07	2.84e-74	radiative	32.4	0.4	0.23
3.98e+33	4.68e+34	1.73e+05	1.34e+08	1.14e+11	5.66e-06	2.84e-74	radiative	39.1	0.4	0.27
3.98e+33	4.68e+34	3.17e+05	1.53e+09	1.1e+11	3.52e-05	2.92e-74	radiative	31.5	0.4	0.222
3.98e+33	4.68e+34	5.38e+05	1.88e+10	1.04e+11	0.000257	2.05e-59	radiative	19.5	0.4	0.203
3.98e+33	4.68e+34	9.36e+05	2.98e+11	9.4e+10	0.00235	6.79e-45	radiative	11.4	0.4	0.206
3.96e+33	4.68e+34	1.76e+06	5.34e+12	7.96e+10	0.0224	3.2e-31	radiative	9.1	0.4	0.233
3.87e+33	4.68e+34	3.53e+06	1.1e+14	6.04e+10	0.231	3.94e-19	radiative	6.05	0.4	0.206
3.23e+33	4.68e+34	6.72e+06	2.96e+15	3.88e+10	3.27	9.32e-10	radiative	2.39	0.4	0.199
1.11e+33	4.46e+34	1.33e+07	4.95e+16	1.93e+10	27.7	0.0614	radiative	1.15	0.4	0.334
3.5e+32	3.63e+34	1.69e+07	9.35e+16	1.22e+10	41.1	10.9	convective	0.918	0.4	0.4
1.71e+32	2.86e+34	1.8e+07	1.11e+17	9.46e+09	45.5	43.1	convective	0.86	0.4	0.4
4.62e+31	1.21e+34	1.91e+07	1.28e+17	6.01e+09	49.6	139	convective	0.815	0.4	0.4
4.87e+30	1.7e+33	1.97e+07	1.38e+17	2.81e+09	52	256	convective	0.793	0.4	0.4
5.03e+29	1.87e+32	1.99e+07	1.41e+17	1.32e+09	52.5	292	convective	0.788	0.4	0.4
5.25e+28	1.98e+31	1.99e+07	1.41e+17	6.2e+08	52.6	301	convective	0.787	0.4	0.4
5.53e+27	2.1e+30	1.99e+07	1.41e+17	2.93e+08	52.7	303	convective	0.787	0.4	0.4
5.71e+26	2.17e+29	1.99e+07	1.42e+17	1.37e+08	52.7	303	convective	0.787	0.4	0.4
5.97e+25	2.26e+28	1.99e+07	1.42e+17	6.47e+07	52.7	303	convective	0.787	0.4	0.4
6.29e+24	2.38e+27	1.99e+07	1.42e+17	3.06e+07	52.7	303	convective	0.787	0.4	0.4
6.49e+23	2.46e+26	1.99e+07	1.42e+17	1.43e+07	52.7	303	convective	0.787	0.4	0.4
6.78e+22	2.57e+25	1.99e+07	1.42e+17	6.75e+06	52.7	303	convective	0.787	0.4	0.4
7.15e+21	2.71e+24	1.99e+07	1.42e+17	3.19e+06	52.7	303	convective	0.787	0.4	0.4
7.38e+20	2.8e+23	1.99e+07	1.42e+17	1.5e+06	52.7	303	convective	0.787	0.4	0.4
7.71e+19	2.92e+22	1.99e+07	1.42e+17	7.04e+05	52.7	303	convective	0.787	0.4	0.4
8.13e+18	3.08e+21	1.99e+07	1.42e+17	3.33e+05	52.7	303	convective	0.787	0.4	0.4
8.39e+17	3.18e+20	1.99e+07	1.42e+17	1.56e+05	52.7	303	convective	0.787	0.4	0.4
8.76e+16	3.32e+19	1.99e+07	1.42e+17	7.35e+04	52.7	303	convective	0.787	0.4	0.4
9.24e+15	3.5e+18	1.99e+07	1.42e+17	3.47e+04	52.7	303	convective	0.787	0.4	0.4
9.54e+14	3.61e+17	1.99e+07	1.42e+17	1.63e+04	52.7	303	convective	0.787	0.4	0.4
9.96e+13	3.77e+16	1.99e+07	1.42e+17	7.67e+03	52.7	303	convective	0.787	0.4	0.4
1.05e+13	3.98e+15	1.99e+07	1.42e+17	3.62e+03	52.7	303	convective	0.787	0.4	0.4
1.08e+12	4.11e+14	1.99e+07	1.42e+17	1.7e+03	52.7	303	convective	0.787	0.4	0.4
1.13e+11	4.29e+13	1.99e+07	1.42e+17	800	52.7	303	convective	0.787	0.4	0.4
1.11e+10	4.21e+12	1.99e+07	1.42e+17	369	52.7	303	convective	0.787	0.4	0.4
1e+10	3.79e+12	1.99e+07	1.42e+17	357	52.7	303	convective	0.787	0.4	0.4
1e+10	3.79e+12	1.99e+07	1.42e+17	357	52.7	303	convective	0.787	0.4	0.4

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